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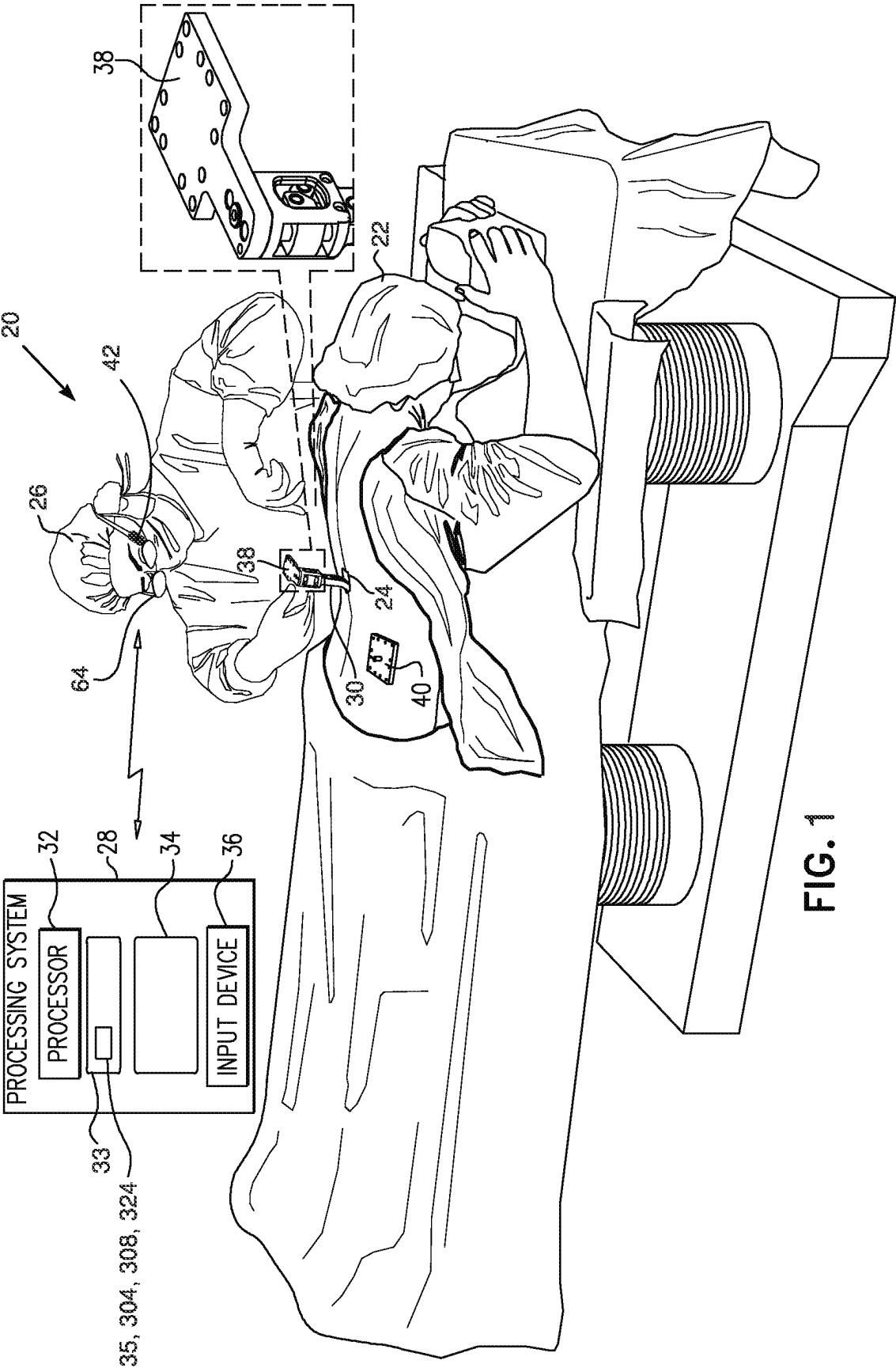
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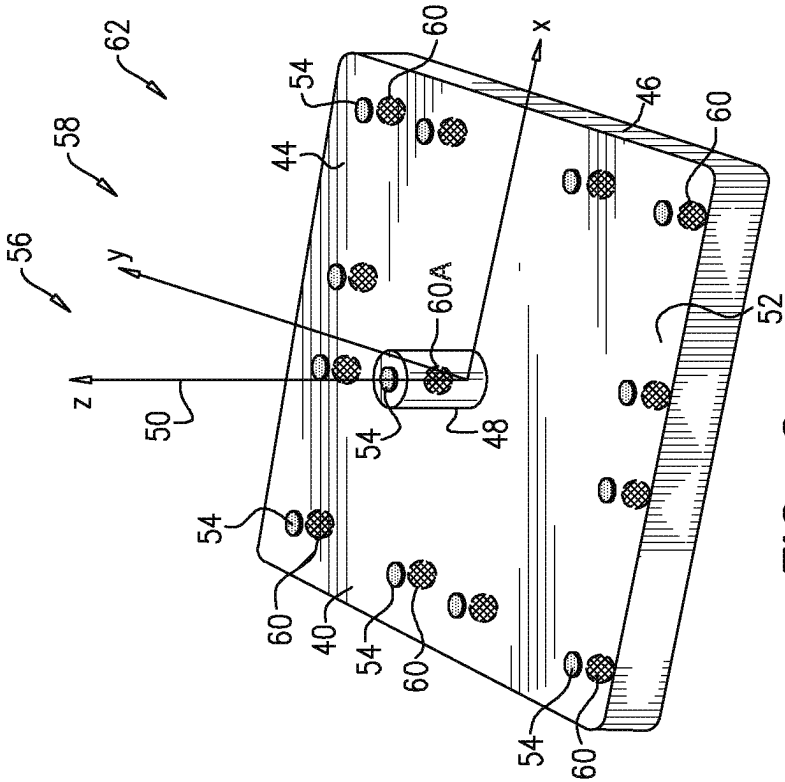


FIG. 2

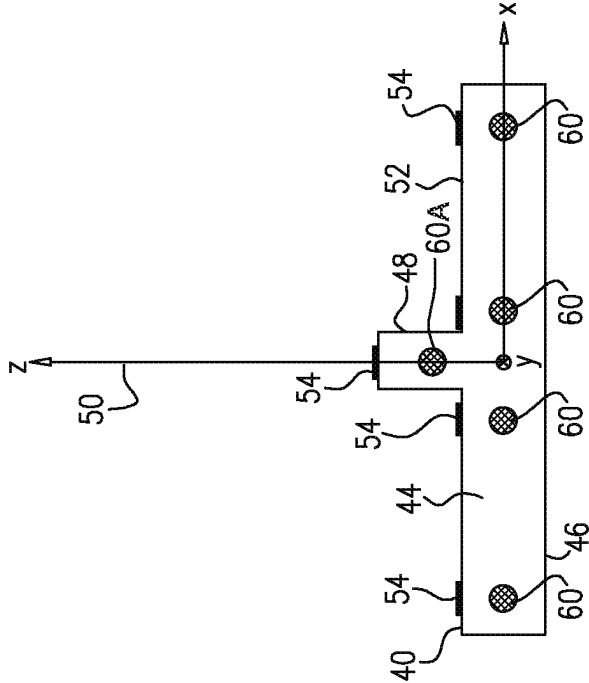


FIG. 3

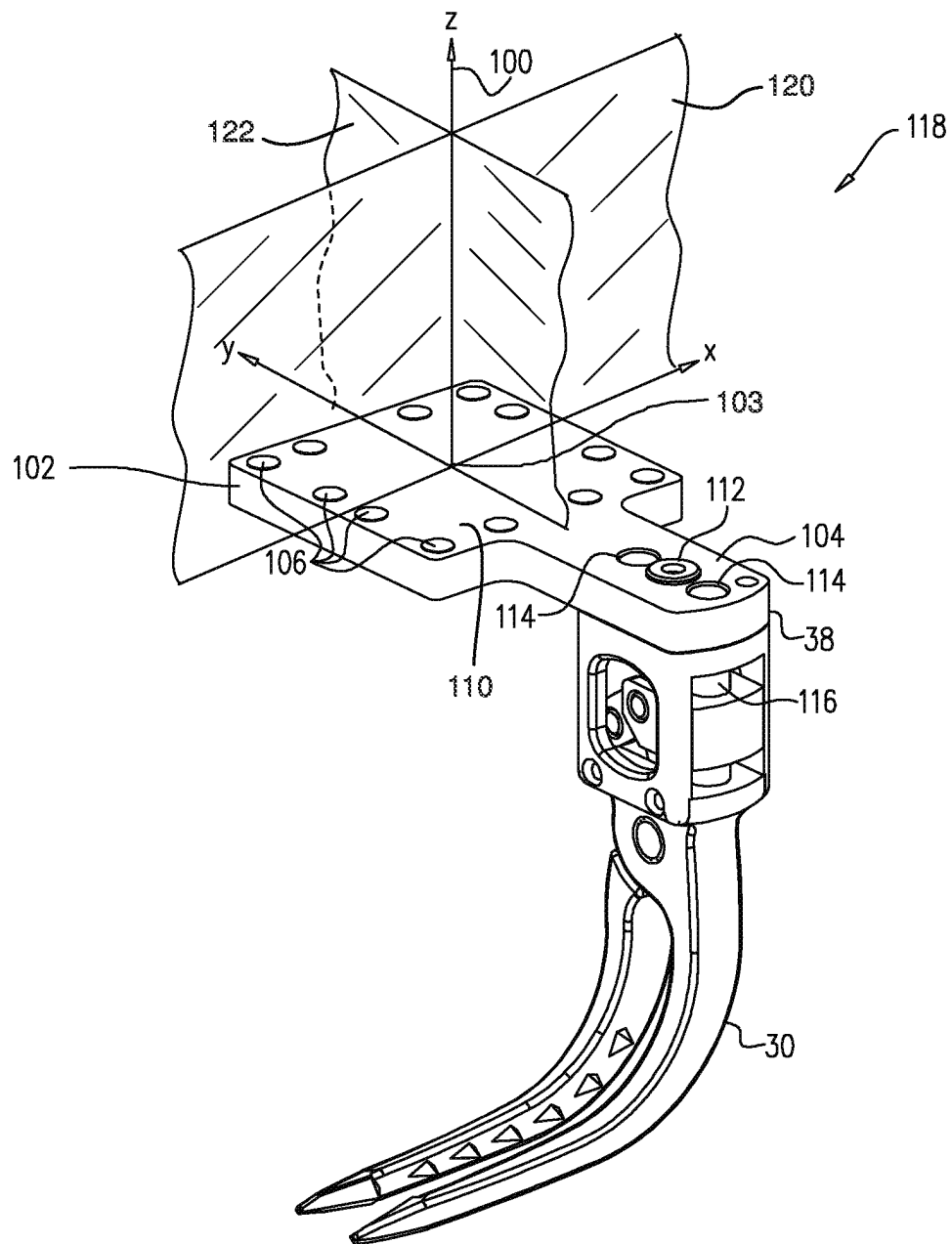


FIG. 4

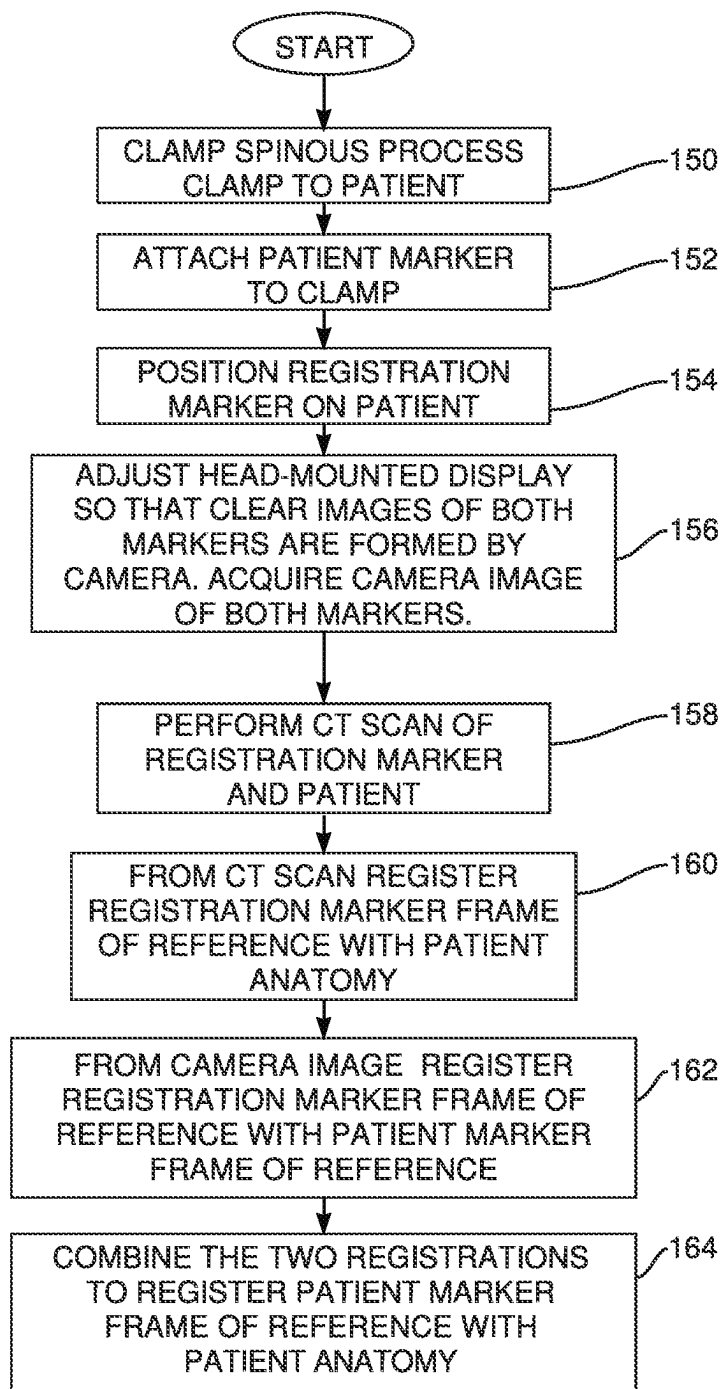
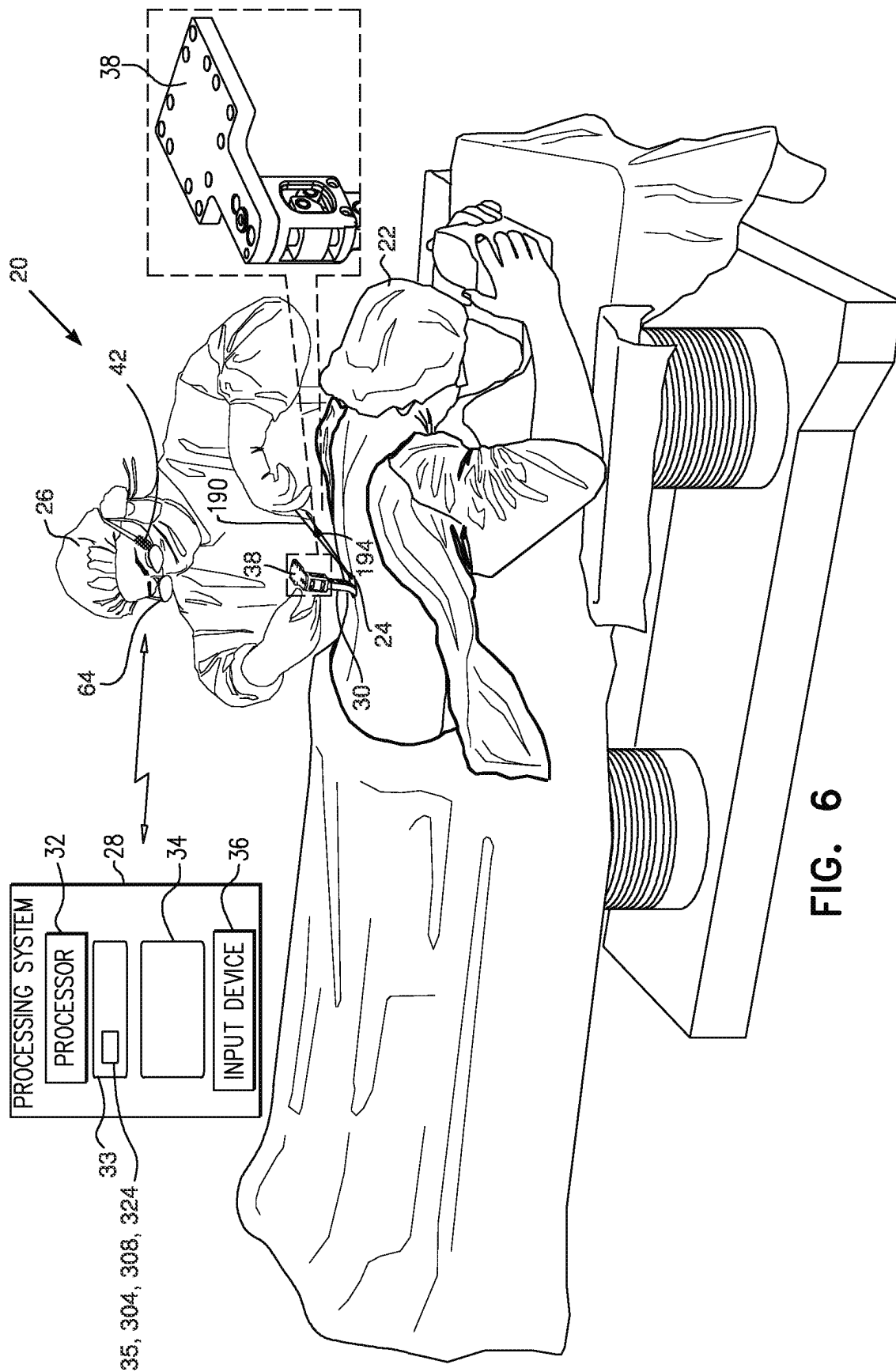


FIG. 5



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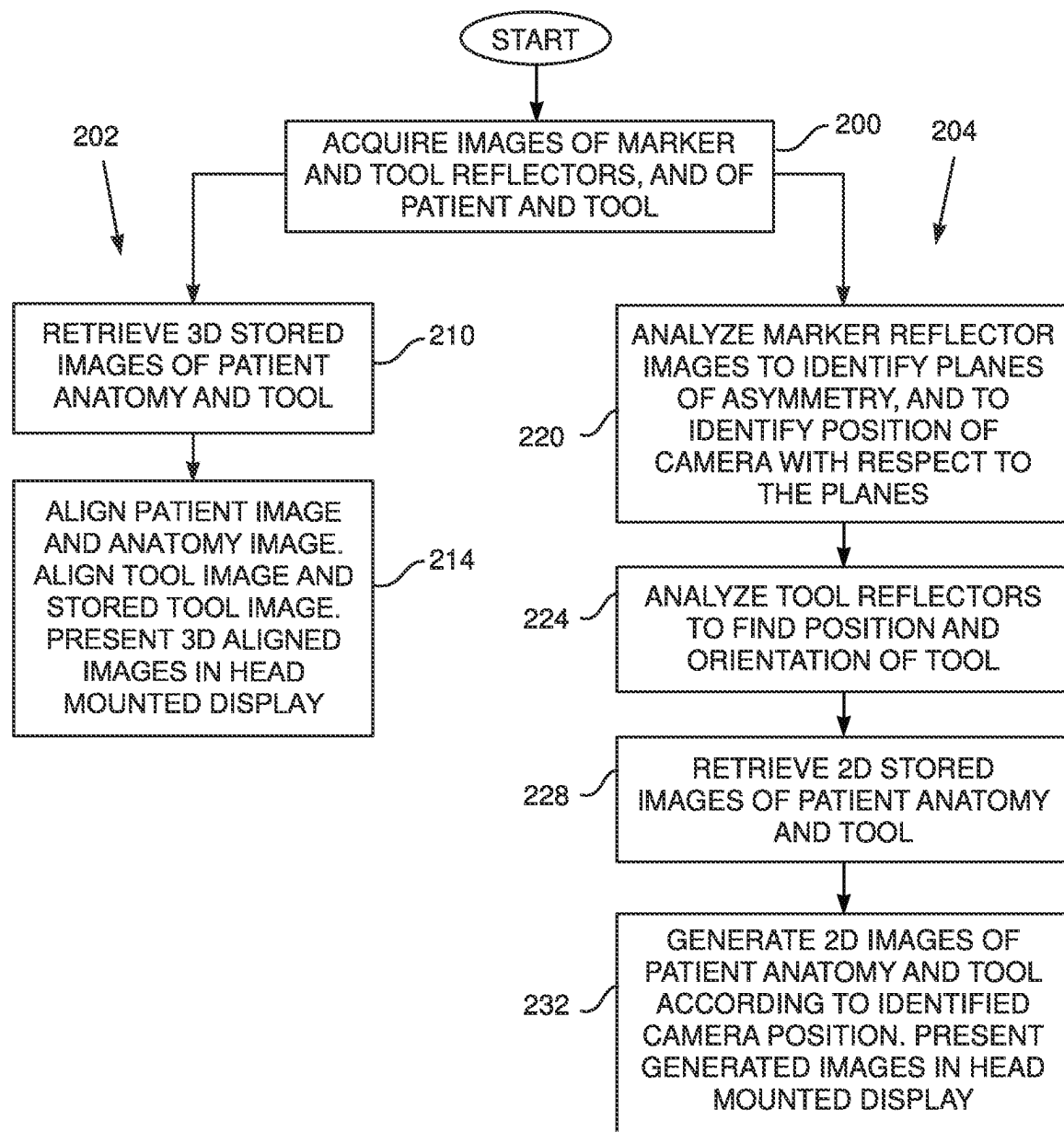


FIG. 7

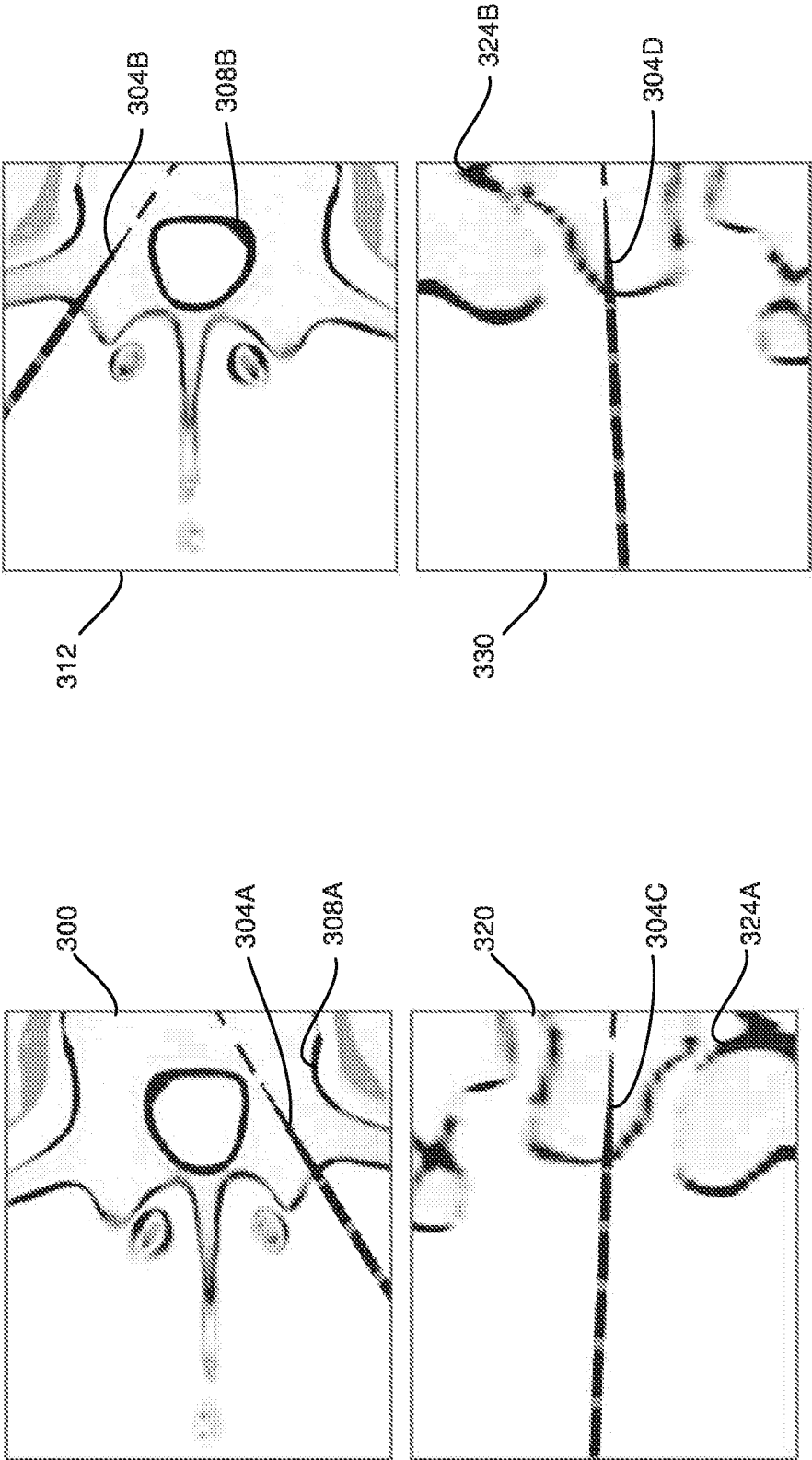


FIG. 8

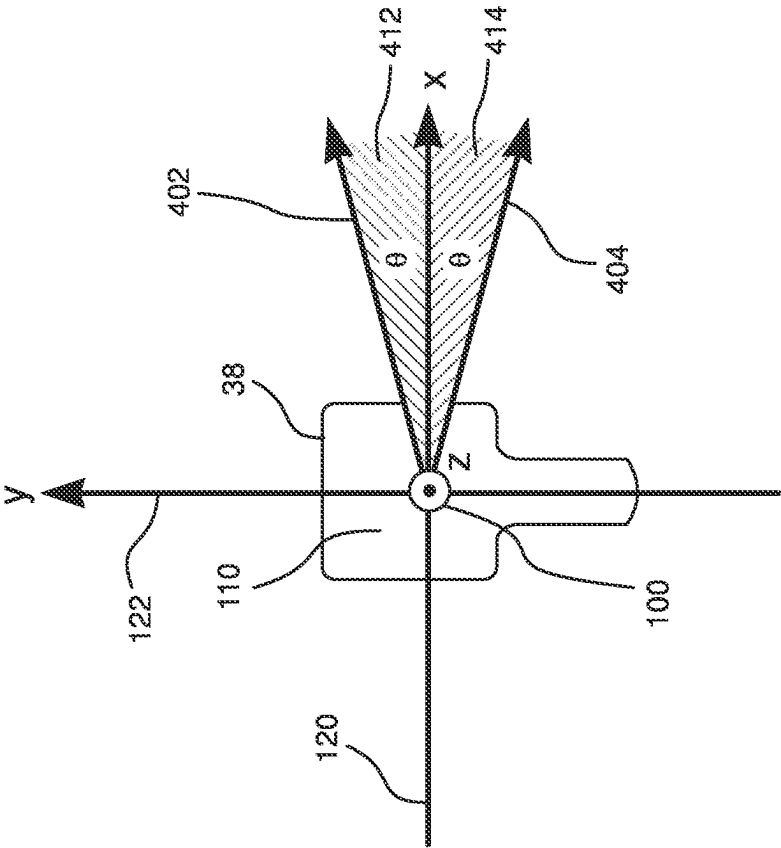
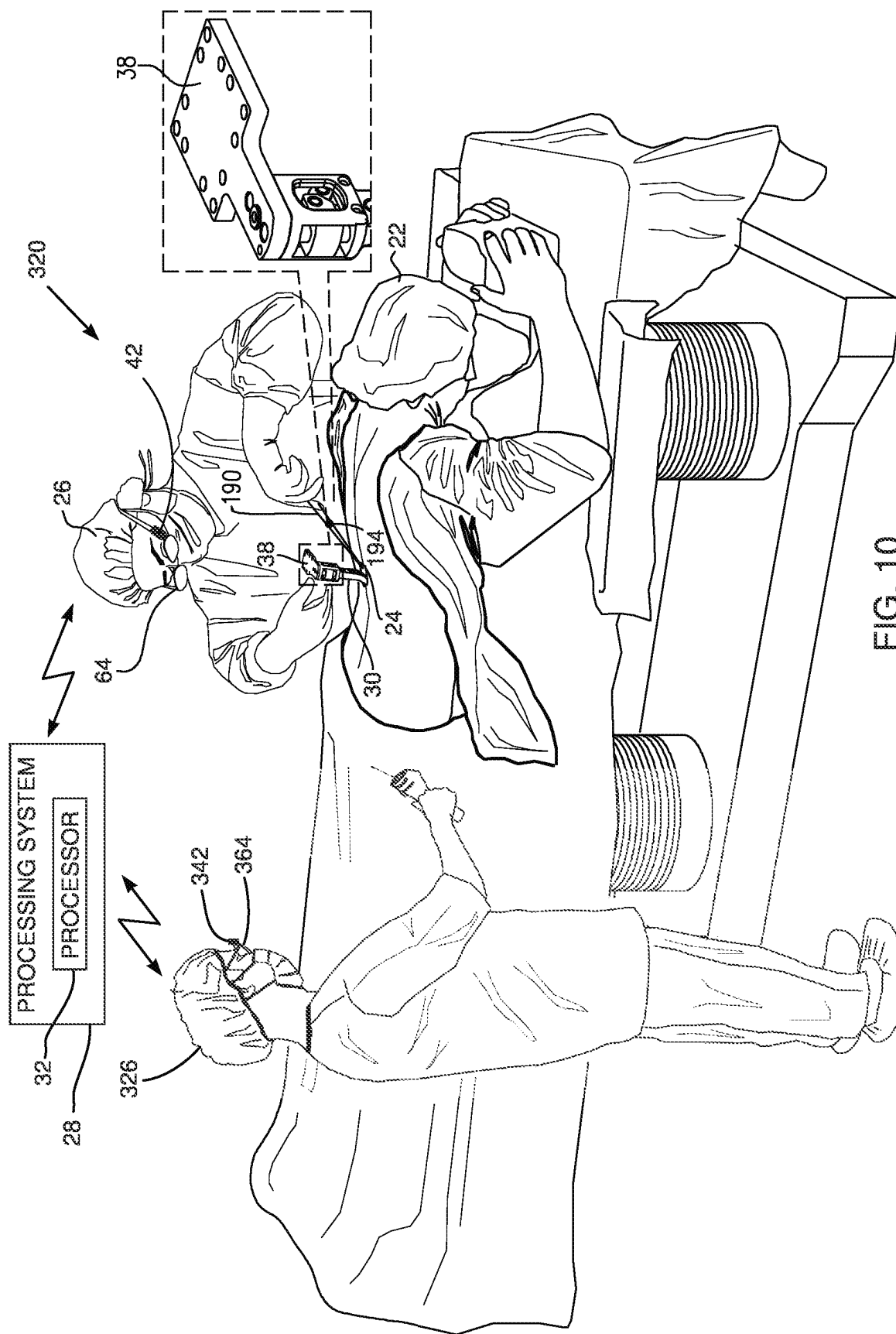


FIG. 9



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MIRRORING IN IMAGE GUIDED SURGERY**CROSS-REFERENCE TO RELATED APPLICATION**

This application is a continuation of U.S. patent application Ser. No. 17/827,710, filed May 29, 2022, which is a continuation of U.S. patent application Ser. No. 16/724,297, filed Dec. 22, 2019 (now U.S. Pat. No. 11,382,712), each of which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

This invention relates generally to an augmented reality system, and specifically to correct image projection when it is used in image guided surgery.

BACKGROUND

Correct imaging is important in image guided surgery, and a number of systems are known in the art for producing correct imaging.

U.S. Pat. Nos. 7,630,753 and 9,757,087, to Simon et al., describe a surgical instrument navigation system that allows a surgeon to invert the three-dimensional perspective of the instrument to match their perspective of the actual instrument.

U.S. Pat. No. 9,538,962, to Hannaford et al., describes a system for providing networked communications. The system includes a plurality of head-mountable devices, each in communication with a control system via a communication network.

U.S. Pat. No. 9,710,968, to Dillavou et al., describes a system for role designation with multiple sources.

U.S. Pat. No. 9,886,552, to Dillavou et al., describes a method for image registration that includes rendering a common field of interest that reflects a presence of a plurality of elements. At least one of the elements is a remote element located remotely from another of the elements.

U.S. Pat. No. 9,940,750, to Dillavou et al., describes a method for role negotiation that can comprise rendering a common field of interest that reflects a presence of a plurality of elements. At least one of the elements is a remote element located remotely from another of the elements.

U.S. Pat. No. 9,959,629, to Dillavou et al., describes a method for managing spatiotemporal uncertainty in image processing. The method can comprise determining motion from a first image to a second image.

U.S. Pat. No. 10,194,131, to Casas, describes a real-time surgery method for displaying a stereoscopic augmented view of a patient from a static or dynamic viewpoint of the surgeon. The method employs real-time three-dimensional surface reconstruction for preoperative and intraoperative image registration.

US Patent Application 2011/0216060, to Weising et al., describes a method for controlling a view of a virtual scene with a portable device. A signal is received and the portable device is synchronized to make the location of the portable device a reference point in a three-dimensional (3D) space.

US Patent Application 2017/0027650, to Merck et al., describes receiving data characterizing a mother video feed acquired by an endoscopic video capture device. The mother video feed can be for characterizing an operative field within a patient.

US Patent Application 2017/0251900, to Hansen et al., describes a depiction system for generating a real time

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correlated depiction of movements of a surgical tool for uses in minimally invasive surgery.

US Patent Application 2017/0367771, to Tako et al., describes a virtual reality surgical navigation method that includes a step of receiving data indicative of a surgeon's current head position, including a direction of view and angle of view of the surgeon.

US Patent Application 2018/0247128, to Alvi et al., describes a system for accessing a surgical dataset including surgical data collected during performance of a surgical procedure. The surgical data can include video data of the surgical procedure.

Documents incorporated by reference in the present patent application are to be considered an integral part of the application except that, to the extent that any terms are defined in these incorporated documents in a manner that conflicts with definitions made explicitly or implicitly in the present specification, only the definitions in the present specification should be considered.

SUMMARY

An embodiment of the present invention provides an imaging system, consisting of:

- a head-mounted display configured to be worn by an operator of the system;
- a marker configured to be attached to a human subject and defining a plane when attached to the human subject, the marker having optically reflective elements disposed on the marker and on opposing sides of the plane in a non-symmetrical arrangement with respect to the plane;
- a memory configured to store a graphical representation of a tool used in a procedure performed by the operator on the human subject, and an image of anatomy of the human subject;
- a camera attached to the display and configured to acquire an input image of the marker and of the tool; and
- a processor configured to analyze the input image so as to identify the plane and to identify a side of the plane wherein the camera is located, and to render to the display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a point of view in the identified side of the plane.

In a disclosed embodiment the plane makes an angle between +20° and -20° with a sagittal plane of the human subject. Alternatively, the plane makes an angle between +20° and -20° with an axial plane of the human subject.

In a further disclosed embodiment the marker has a two-dimensional surface which makes an angle between +20° and -20° with a frontal plane of the human subject.

In a yet further disclosed embodiment the marker defines a further plane and the optically reflective elements are disposed on opposing sides of the further plane in a non-symmetrical arrangement with respect to the further plane, and the processor is configured to analyze the input image so as to identify the further plane and to identify a side of the further plane wherein the camera is located, and to render to the display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a point of view in the identified side of the further plane. Typically, the plane and the further plane are orthogonal to each other.

In an alternative embodiment the camera is located at a vertical height above the marker, and the processor is configured:

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to ascertain the vertical height in response to the acquired input image of the marker;
 to calculate a pair of planes, each of the pair having a preset acute angle to the identified plane and defining a first acute-angled wedge region and a second acute-angled wedge region to the identified plane; and
 when the display moves so that the point of view crosses the first acute-angled wedge region and the second acute-angled wedge region, or begins within the first acute-angled wedge region and crosses the second acute-angled wedge region, while the camera remains at the vertical height, to render to the display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from the point of view of a region opposite the identified side.

Typically the preset acute angle is less than or equal to 10°.

In a further alternative embodiment the camera is located at a vertical height above the marker, and the processor is configured:

to ascertain the vertical height in response to the acquired input image of the marker; and
 when the display moves so that the vertical height changes, to render unchanged to the display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon.

There is further provided, according to an embodiment of the present invention, an imaging system, consisting of:

a first head-mounted display configured to be worn by a first operator of the system;
 a second head-mounted display configured to be worn by a second operator of the system;
 a marker configured to be attached to a human subject and defining a plane when attached to the human subject, the marker having optically reflective elements disposed on the marker and on opposing sides of the plane in a non-symmetrical arrangement with respect to the plane;
 a memory configured to store a graphical representation of a tool used in a procedure performed by the first operator on the human subject, and an image of anatomy of the human subject;
 a first camera attached to the first display and configured to acquire a first input image of the marker and of the tool;
 a second camera attached to the second display and configured to acquire a second input image of the marker and of the tool; and
 a processor configured to:

analyze the first input image so as to identify the plane and to identify a first side of the plane wherein the first camera is located, and to render to the first display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a first point of view in the identified first side of the plane, and

analyze the second input image so as to identify the plane and to identify a second side of the plane wherein the second camera is located, and to render to the second display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a second point of view in the identified second side of the plane.

There is further provided, according to an embodiment of the present invention, a method, consisting of:

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providing a head-mounted display configured to be worn by an operator of an imaging system;
 attaching a marker to a human subject, the marker defining a plane when attached, the marker having optically reflective elements disposed on the marker and on opposing sides of the plane in a non-symmetrical arrangement with respect to the plane;
 storing in a memory a graphical representation of a tool used in a procedure performed by the operator on the human subject, and storing an image of anatomy of the human subject in the memory;
 attaching a camera to the display;
 acquiring an input image of the marker and of the tool with the camera; and
 analyzing the input image so as to identify the plane and to identify a side of the plane wherein the camera is located, and to render to the display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a point of view in the identified side of the plane.

The present disclosure will be more fully understood from the following detailed description of the embodiments thereof, taken together with the drawings. A brief description of the drawings follows.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic illustration of an initial preparatory stage of a medical procedure, according to an embodiment of the present invention;

FIGS. 2, 3, and 4 are schematic depictions of entities used in the initial stage, according to an embodiment of the present invention;

FIG. 5 is a flowchart of steps performed to register a patient marker with the anatomy of a patient during the initial preparatory stage;

FIG. 6 is a schematic illustration of a subsequent stage of the procedure, according to an embodiment of the present invention;

FIG. 7 is a flowchart of steps performed during the subsequent stage, according to an embodiment of the present invention;

FIG. 8 shows schematic figures illustrating images generated in the subsequent stage, according to an embodiment of the present invention;

FIG. 9 is a schematic top-down view of a surface of a marker used in the procedure; and

FIG. 10 is a schematic illustration of the subsequent stage of the procedure when there are two operators for the procedure, according to an embodiment of the present invention.

DETAILED DESCRIPTION OF EMBODIMENTS

Overview

A head-mounted display, for a medical procedure that implements an imaging system, such as an augmented reality system, in the display, typically needs to access stored computerized tomography (CT) files of the anatomy of a human subject. The display is worn by an operator of the system, and the accessed files are presented to the operator as scanned planes of the subject in the display. However, for the presentation to be correctly oriented, it is necessary to know the position of the operator with respect to the subject.

Embodiments of the present invention provide an imaging system that determines the operator position automatically,

and so displays an image of the patient anatomy, and of a tool used in the procedure, automatically.

In addition to a head-mounted display (HMD) that is worn by an operator of the system, the system comprises a marker that is attached to the human subject. The marker defines a plane of asymmetry when attached to the human subject, since the marker has optically reflective elements disposed on the marker and on opposing sides of the plane in a non-symmetrical arrangement with respect to the plane. The plane of asymmetry is typically approximately parallel to one of the main anatomical planes of the human subject.

In the imaging system a memory stores a graphical representation of a tool used in the procedure performed by the operator, and the memory also stores an image of the anatomy of the human subject. A camera is attached to the HMD, and acquires an input image of the marker and of the tool. A processor analyzes the input image so as to identify the plane and to identify a side of the plane wherein the camera is located. The processor then renders to the display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a point of view in the identified side of the plane.

DETAILED DESCRIPTION

In the following, all directional references (e.g., upper, lower, upward, downward, left, right, top, bottom, above, below, vertical, and horizontal) are only used for identification purposes to aid the reader's understanding of the present invention, and do not create limitations, particularly as to the position, orientation, or use of embodiments of the invention.

In the description, like elements in the drawings are identified by like numerals, and like elements are differentiated as necessary by appending a letter to the identifying numeral.

Reference is now made to FIGS. 1, 2, 3, and 4, which are diagrams according to an embodiment of the present invention. FIG. 1 is a schematic illustration of an initial preparatory stage of a medical procedure using an imaging system 20, and FIGS. 2, 3, and 4 are schematic depictions of entities used in the initial stage. The medical procedure exemplified here is performed on the back of a human subject 22, herein also termed patient 22, and during the initial stage of the procedure an operator 26 of system also herein termed medical professional 26 makes an incision 24 into the patient's back. The professional inserts a spinous process clamp 30 into the incision, so that opposing jaws of the clamp are located on opposite sides of the spinous processes. The professional then slides the clamp over the vertebral laminae, and adjusts the clamp to grip one or more spinous processes, selected by the professional, of the patient. Clamp 30 is described below with reference to FIG. 4, and a clamp such as clamp 30 is described in more detail in U.S. Patent Application 2019/0175228 which is incorporated herein by reference.

Clamp 30 acts as a support for a patient marker 38, which is attached rigidly to the clamp. During substantially all of the procedure, i.e., during the initial, as well as the subsequent stages, patient marker 38 is used as a fiducial for patient 30, since because of its rigid connection to the patient, any movement of the patient is reflected in a corresponding motion of the patient marker. In order to operate as such a fiducial, in embodiments of the present invention, in the initial stage of the procedure marker 38 is registered with the anatomy of patient 30, herein assumed to comprise the skeleton of the patient, as is described herein.

During the procedure medical professional 26 wears a head-mounted display (HMD) 64 which is configured to present stored images, that are aligned with patient 22, to professional 26. HMD 64 is described further below.

As is also described below, in serving as a fiducial, marker 38 performs two functions: a first function wherein the marker is used to maintain registration between frames of reference of the head-mounted display and the patient's anatomy, and a second function wherein the marker is used to ascertain where the medical professional is located with respect to the patient. Thus, for the second function, the marker provides a location of the medical professional as being on a left side or a right side of the patient, or on an upper side or a lower side of the patient.

An augmented reality head-mounted display such as HMD 64 is described in more detail in U.S. Patent Application 2017/0178375 which is incorporated herein by reference.

During the initial stage of the procedure, a registration marker 40 is placed on the patient's back, and is used to implement the registration of patient marker 38 with the anatomy of patient 30. In contrast to patient marker 38, registration marker 40 is typically only used during the initial stage of the procedure, i.e., for the registration of the patient marker 38, and once the registration has been performed, for the subsequent procedure stages the registration marker may be removed from the patient's back. As will be apparent from the following description, only registration marker 40 is subject to fluoroscopy, and patient marker 38 is not subject to fluoroscopy.

Also during the initial stage of the procedure, a camera 42, fixedly attached to head-mounted display 64, is used to image the registration marker and the patient marker. Camera 42 typically operates in the visible and/or near-visible spectrum, i.e., at wavelengths of approximately 300 nm-900 nm.

A processing system 28 is coupled, by cables and/or wirelessly, to camera 42. System 28 comprises a computer processor 32, a memory 33 comprising stored images 35 that include images 304, 308, and 324, described below, a screen 34, and an input device 36 such as a pointing device. The system is configured to analyze the images acquired by the camera, as is described further below. Other functions of system 28 are also described below.

In order to operate, HMD 64 is coupled to processor 32 of system 28, or alternatively HMD 64 has its own dedicated processor which performs similar functions to those performed by processor 32. When HMD 64 is operative it presents stored images, that are aligned with patient 22, to professional 26.

FIGS. 2 and 3 are respectively schematic perspective and cross-sectional views of registration marker 40, which is assumed to define a registration marker frame of reference 50, herein assumed to comprise an orthogonal set of xyz axes. Marker 40 is formed from a solid substrate 44, which is opaque to light in the visible and near-visible spectrum, and which is transparent to fluoroscopic radiation. Substrate 44 is typically formed from a hard plastic, such as polycarbonate, but any other solid material which is opaque to light and transparent to fluoroscopic radiation may be used in embodiments of the present invention.

In the illustrated embodiment of marker 40, substrate 44 is formed as a rectangular parallelepiped 46, upon which is mounted a pillar 48.

A plurality of optically reflective, but radiotransparent, discrete elements 54 are disposed on substrate 44. Elements 54 are hereinbelow, by way of example, assumed to com-

prise discs, and are also referred to herein as discs **54**. It is understood that said optically reflective and radiotransparent elements may be of different shapes and/or sizes.

Some of the plurality of discs **54** are fixedly attached, typically by cementing, to a two-dimensional (2D) surface **52** of parallelepiped **46**. These discs **54** are formed in a generally rectangular 2D pattern on surface **52**. In addition, an optically reflective disc **54** is also cemented onto pillar **48**, so that there is in totality a three-dimensional (3D) array of discs **54** disposed on the substrate. The 3D array of discs **54** are distributed on 2D surface **52**, and on pillar **48**, so that when marker **40** is illuminated and imaged by camera **50** the discs are easily distinguished from substrate **44**. Furthermore, as explained in more detail below, the arrangement of discs **54** are configured to enable processor **32** to unambiguously determine the orientation and position of frame of reference **50** from the marker image.

The distributed discs **54** are herein assumed to comprise an optical component **56** of marker **40** that forms an optical pattern **58** for the marker. In a particular aspect of the invention optical pattern **58**, comprising the distribution of discs **54**, is implemented so that the pattern has no axis of symmetry and no plane of symmetry. The absence of both an axis and a plane of symmetry in the pattern ensures that the unambiguous determination of the orientation and position of the frame of reference of marker **40** is possible from the marker image for multiple different orientations and positions of the marker, the positions being typically within a region approximately 20 cm from the patient marker.

The description above of optical pattern **58** assumes that discs **54** are configured in three dimensions. However, as long as the pattern has no axis of symmetry and no plane of symmetry, the discs forming the pattern may be arranged in only two dimensions, for example, absent the disc on pillar **48**. Thus, pattern **58** may be formed in at least two dimensions, i.e., in the case of discs **54**, as a two-dimensional array of the discs or as a three-dimensional array of the discs.

It will be understood that the requirement for discs **54** to be arranged to form a pattern having an absence of both an axis and a plane of symmetry may be achieved using discs of substantially the same size and shape, wherein locations of the discs are selected so that the locations are arranged to have the absence of both an axis and a plane of symmetry. The described pattern is hereinbelow referred to as a unique optical pattern.

Alternatively, the unique optical pattern may be achieved using discs of different sizes and/or shapes. In this case, the locations of the discs may also satisfy the requirement, but this is not a necessity.

A multiplicity of radiopaque elements **60** are disposed in substrate **44** by being embedded in a distribution within parallelepiped **46**. The distribution of elements **60** is arranged in a two dimensional radiopaque pattern **62** such that, as for the pattern of discs **54**, the radiopaque pattern has no axis of symmetry and no plane of symmetry. Because substrate **44** is radiotransparent, and because of the absence of both an axis and a plane of symmetry in radiopaque pattern **62**, a fluoroscopic, typically computerized tomography (CT), scan of the radiopaque elements of marker **40** enables the orientation and position of frame of reference **50** to be unambiguously determined by processor **32** from the fluoroscopic scan. In one embodiment elements **60** comprise spheres which are distributed in a 2D generally rectangular 2D pattern that is substantially the same as the rectangular pattern of discs **54** on surface **52**.

The description above of elements **60** assumes that they are arranged in a radiopaque pattern of two dimensions.

However, as long as the pattern has no axis of symmetry and no plane of symmetry, the elements forming the pattern may also be arranged in three dimensions, for example, by incorporation of a radiopaque element **60A**, substantially similar to elements **60**, in pillar **48**. Thus, pattern **62** may also be formed in at least two dimensions, i.e., in the case of elements **60** and **60A**, as a two-dimensional array of elements **60** or as a three-dimensional array of elements **60** and **60A**.

As for discs **54**, it will be understood that the requirement for elements **60** to be arranged to form a pattern having an absence of both an axis and a plane of symmetry may be achieved using elements of substantially the same size and shape, wherein locations of the elements are selected so that the locations are arranged to have the absence of both an axis and a plane of symmetry. The described pattern is hereinbelow referred to as a unique radiopaque pattern.

Alternatively, the unique radiopaque pattern may be achieved using elements of different sizes and/or shapes. In this case, the locations of the elements may also satisfy the requirement, but this is not a necessity.

The X-ray wavelengths of the CT scan are assumed to be in a range of 0.01-10 nm.

The above description of marker **40** assumes that discs **54** and elements **60** have different functionalities—the discs being optically reflective and radiotransparent, and the elements being radiopaque. In an alternative embodiment of marker **40** at least some of discs **54** are configured to have dual functionality by being optically reflective and radiopaque. As for the embodiment described above, in the alternative embodiment discs **54** are configured and distributed on substrate **44** so that an optical image of marker **40** provides an unambiguous determination of the orientation and position of frame of reference **50**, and a fluoroscopic scan of the marker also provides an unambiguous determination of the orientation and position of the frame of reference.

The physical construction of the illustrated embodiment of marker **40**, as a pillar attached to a rectangular parallelepiped, comprising an array of discs **54** and an array of elements **60**, is but one example of possible physical constructions of the marker that enables an unambiguous determination of the marker's position and orientation from a camera image and from a fluoroscopic scan. In a disclosed embodiment, rather than marker **40** comprising pillar **48** mounted on substrate **44**, an indentation (in place of the pillar) is formed within the substrate, and a disc **54** is located on a surface of the indentation.

Other suitable constructions for marker **40** are also considered to be within the scope of the present invention.

For example, the substrate of marker **40**, rather than being formed from a parallelepiped with a pillar or an indentation, may be formed as substantially any conveniently shaped solid object that is opaque to light in the visible and near-visible spectrum and which is transparent to fluoroscopic radiation.

In addition, rather than the optical component of marker **40** being comprised of a plurality of discs **54** arranged in a particular pattern, the component may comprise any array or pattern of optical elements that is attached to the substrate, that is diffusely and/or specularly reflective, and that is configured to have the absence of axes and planes of symmetry described above, so that when imaged in visible or near-visible light an unambiguous determination of the marker's position and orientation may be made.

Referring to FIG. 4, patient marker **38** is assumed to define a patient marker frame of reference **100**, assumed to

comprise an orthogonal set of xyz axes. In the embodiment illustrated in FIG. 4 marker 38 comprises a rectangular parallelepiped substrate 102 to which is attached a tongue 104 used to fixedly connect the substrate to clamp 30. A center 103 of an upper surface of substrate 102 acts as an origin of the xyz axes.

The connection to clamp 30 is by a removable screw 112, and the patient marker connects in a predetermined fixed spatial relationship to the clamp using holes 114 which align with studs 116 of the clamp. Substrate 102 comprises a solid opaque material, and may be formed from any convenient material such as polyimide plastic.

A plurality of optically reflective discs 106, generally similar to discs 54, are attached, typically by cementing, to an upper 2D surface 110 of substrate 102. Discs 106, also referred to herein as reflectors 106, are formed in a generally rectangular 2D pattern on surface 110. Discs 106 are distributed so that when illuminated and imaged by camera 42 they are easily distinguished from substrate 102.

In addition, discs 106 are distributed with respect to an xz plane 120 and a yz plane 122 through origin 103. xz plane 120 and yz plane 122 are planes of asymmetry. Thus, discs 106 are arranged non-symmetrically with respect to xz plane 120, so that the distribution of the discs on one side of plane 120 do not mirror (through the plane) the discs on the opposing side of the plane. In addition, discs 106 are arranged non-symmetrically with respect to yz plane 122, so that the distribution of the discs on one side of plane 122 do not mirror the discs on the opposing side of the plane.

In FIG. 4 discs 106 are shown as being distributed on sides of a rectangle, however, it will be understood that this is but one example for the positioning of the discs on surface 110. Other distributions of discs 106, providing that they define planes of asymmetry as described above, are also assumed to be comprised within the scope of the present invention.

Furthermore, it will be appreciated that the physical construction of patient marker 38 described above is by way of example. Thus, embodiments of the present invention comprise any patient marker formed of any conveniently shaped solid opaque substrate to which is attached an optical pattern, the pattern defining planes of asymmetry as described above.

FIG. 5 is a flowchart of steps performed to register patient marker 38 with the anatomy of patient 22 during the initial preparatory stage of a medical procedure illustrated in FIG. 1, according to an embodiment of the present invention. While the following description assumes, for simplicity, a CT scan, other types of fluoroscopic imaging are also considered to be within the scope of the present invention.

In an initial step 150, medical professional 26 makes an incision in the back of patient 22, inserts spinous clamp 30 into the patient, and then clamps the clamp to one or more of the processes of the patient.

In a patient marker step 152, the medical professional attaches patient marker 38 to spinous clamp 30, ensuring that the marker is rigidly attached to the clamp. Marker 38 is attached to clamp 30 so that surface 110, corresponding to the xy plane of the xyz axes, is approximately parallel to a frontal plane of patient 22, xz plane of asymmetry 120 is approximately parallel to a sagittal plane of the patient, and so that yz plane of asymmetry 122 is approximately parallel to an axial plane of the patient. As used herein, the term "approximately parallel" as applied to two planes indicates that the planes subtend an angle within a range of $\pm 20^\circ$ to each other.

In a registration marker step 154, the professional places registration marker 40 on the skin of the back of the patient, typically as close to the patient's spine as is convenient.

In a camera step 156, professional 26 adjusts his/her position so that camera 42, attached to head-mounted display 64 images the registration marker and the patient marker. Professional 26 adjusts their position so that the images formed by camera 42 of the registration marker and of the patient marker are clear images, i.e., that neither marker occludes the other. Typically processor 32 of processing system 28 is configured to verify the acceptability of the two marker images, and if necessary the professional may use and communicate with system 28 to adjust, in an iterative manner, their position and/or that of the registration marker until system 28 provides an indication to the professional that acceptable images are being generated.

Once acceptable images are being generated, a camera image of the two markers is acquired, and is provided to processing system 28.

In a fluoroscopic scan step 158, a CT scan of patient 22, in the vicinity of marker 40 is performed, and processing system 28 acquires the scan. The scan may be performed by inserting patient 22 into a CT scanning system so that marker 40 is scanned. The insertion may be implemented by bringing the CT scanning system to patient 22, or by transporting the patient to the system. In either case, marker 40 remains in the marker's position of step 156.

In a scan analysis step 160, processor 32 analysis the CT scan acquired in step 158, the scan comprising an image of radiopaque elements 60 and of the anatomy of patient 22.

From the acquired image, processor 32 calculates the position and orientation of registration marker frame of reference 50, and registers the frame of reference with the anatomy of the patient. The registration typically comprises a set of vectors P between selected points on registration marker 40 and selected vertebrae of patient 22. In one embodiment, the registration comprises using a 4x4 homogenous transformation, comprising a 3x3 rotation and a 1x3 translation, that transforms a point in the space of patient 22 to a point in registration marker frame of reference 50.

In a camera image analysis step 162, processor 32 analyzes the camera image of patient marker 38 and registration marker 40 acquired in step 156. From the acquired image, processor 32 calculates the position and orientation of registration marker frame of reference 50, and the position and orientation of patient marker frame of reference 100. Once the processor has calculated the positions and orientations of the two frames of reference, it formulates a registration of the two frames of reference as a set of vectors Q describing the transformation of the registration marker frame of reference to the patient marker frame of reference.

In a concluding analysis step 164, the processor adds the two sets of vectors found in steps 160 and 162 to formulate a registration set of vectors R between the patient marker frame of reference 36 and the patient anatomy, as shown in equation (1):

$$R = P + Q \quad (1)$$

FIG. 6 illustrates a subsequent stage of the medical procedure, FIG. 7 is a flowchart of steps performed during the subsequent stage, and FIG. 8 shows schematic figures illustrating images generated in the subsequent stage, according to an embodiment of the present invention. In the subsequent stage registration marker 40 has been removed from the back of patient 22, and medical professional 26 operates on the patient using a surgical tool 190. The tool is

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tracked by the HMD processor, by having identifying reflectors **194**, generally similar to reflectors **106**, attached to the tool.

In an initial step **200** of the flowchart of FIG. 7, the HMD projects visible or invisible light to patient marker **38** and tool **190**. Camera **42** acquires images of reflectors **106** of the marker, of reflectors **194** of tool **190** and of patient **22** and tool **190**.

The flowchart then branches into two paths, a first path **202** and a second path **204**. Processor **32** implements steps of both paths substantially simultaneously.

In first path **202**, in a three-dimensional (3D) image retrieval step **210**, processor **32** retrieves a 3D stored patient anatomy image of patient **22**, typically comprising a CT image of the patient, from stored images **35**. The processor also retrieves a stored virtual image, also herein termed a stored representation, of tool **190** from the stored images.

In a 3D image presentation step **214**, the processor presents aligned 3D images of the patient anatomy and of the virtual tool image in the head mounted display.

The position of the virtual tool image is determined from reflectors **194**. In order to ensure that the anatomy image and the virtual tool image, projected by the display, align with the anatomy of patient **22** and with the actual tool image, the processor determines the position and orientation of frame of reference **100** of the patient marker from the acquired images of reflectors **106**. The processor applies the registration set of vectors **R**, found in step **164** of the flowchart of FIG. 5, to the position and orientation of the marker frame of reference, so as to effect the alignment.

In second path **204**, in a plane identification step **220**, processor **32** analyzes the images of reflectors **106** acquired by camera **42** to identify the position and orientation of xz plane of asymmetry **120** and yz plane of asymmetry **122**. From the images the processor also calculates and stores the height of camera **42** above the xy plane.

From the identified positions and orientations of the planes the processor determines on which side of the planes camera **42** resides. Each plane has two sides, and it will be understood that the two planes divide the volume around marker **38** into four regions, the camera residing in one of four regions.

In a tool reflector step **224** the processor analyzes the images of reflectors **194** to find the position and orientation of tool **190**.

In an image retrieval step **228** the processor retrieves a stored virtual image of the tool. The processor also retrieves, from the stored 2D images, images of the patient anatomy at the tool position, and parallel to the axial and sagittal planes of the patient.

In an image presentation step **232**, the processor uses the retrieved images to generate a combined image of the patient anatomy with a representation of the tool superimposed on the patient anatomy, from a point of view of the camera, i.e., from a point of view in the plane sides identified in step **220**.

The processor presents the combined image in HMD **64** for viewing by professional **26**.

By presenting images in HMD **64** according to the point of view of camera **42**, embodiments of the present invention present correctly oriented images to operator **26**, who is wearing the HMD. It will also be understood that the correct orientation is determined according to the position of the operator **26** with respect to the patient, i.e., whether the operator is to the left or right of the patient, and whether the operator is on a lower or upper side of the patient.

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FIG. 8 shows schematic illustrations of images generated in step **232**, according to an embodiment of the present invention.

A diagram **300** illustrates an image **304A** of tool **190** superimposed on an image **308A** of the patient anatomy, from a point of view in a left side of a sagittal plane of patient **22**, and a diagram **312** illustrates an image **304B** of tool **190** superimposed on an image **308B** of the patient anatomy, from a point of view in a right side of the patient sagittal plane. The two diagrams are mirror images of each other, and use a stored image **304** of tool **190**. The two diagrams also use a stored image **308** of the patient anatomy that is parallel to the patient sagittal plane at an identified position of tool **190**.

A diagram **320** illustrates an image **304C** of tool **190** superimposed on an image **324A** of the patient anatomy, from a point of view in a lower side of an axial plane of patient **22**, and a diagram **330** illustrates an image **304D** of tool **190** superimposed on an image **324B** of the patient anatomy, from a point of view in an upper side of the patient axial plane. As for diagrams **300**, **312**, the two diagrams **320**, **330** are mirror images of each other, and use stored image **304** of tool **190**. Diagrams **320**, **330** use a stored image **324** of the patient anatomy that is parallel to the patient axial plane at the identified position of tool **190**.

Returning to the flowchart of FIG. 7, it will be appreciated that professional **26** may select which images, referred to in steps **214** and **232**, are rendered for viewing in the head-mounted display. Thus the professional may view either the 3D images of step **214**, or the 2D images of step **232**, or both images simultaneously.

FIG. 9 is a schematic top-down view of surface **110** of marker **38**, showing the x, y, and z axes of the marker, as well as xz plane **120** and yz plane **122**.

As operator **26** moves from one side of xz plane **120** to the other side, then following on from step **232** of the flowchart of FIG. 7 together with the diagrams of FIG. 8, the images presented to the operator are mirror images of each other. The mirroring is also true when the operator moves from one side of yz plane **122** to the other side.

A disclosed embodiment of the present invention places a limitation on the mirroring described above when moving from one side of a plane to another, in order to reduce jitter in the presented images when the operator is close to the plane. In order to reduce jitter, the processor constructs transition regions around xz plane **120** and other transition regions around yz plane **122**. The following description is for the transition region around xz plane **120** and to the right of yz plane **122**.

Processor **32** constructs a first plane **402** containing and terminating at the z axis, and at an angle $+ \theta$ from xz plane **120**, and a second plane **404** containing and terminating at the z axis, and at $- \theta$ from xz plane **120**. In one embodiment $0 \leq \theta \leq 10^\circ$. The two planes form respective wedge-shaped regions **412**, **414** with xz plane **120**, and these two wedge-shaped regions comprise the transition region around xz plane **120** and to the right of yz plane **122**.

If the movement across xz plane **120** includes both wedge-shaped regions being crossed, by the HMD and the attached camera of the operator, or begins from within one of the wedge-shaped regions and crosses the other one, then the mirroring as described above is implemented.

However, if the movement across the xz plane does not comply with the movements above, e.g., the movement only crosses one wedge-shaped region and stops in the other region, or only moves between wedge-shaped regions, then no mirroring is implemented.

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For a transition region around xz plane **120** and to the left of yz plane **122**, the processor constructs two planes making angles $\pm\theta$ with the xz plane, generally similar to planes **402** and **404**, so as to form two more wedge-shaped regions terminating at the z axis and to the left of the yz plane.

The processor constructs the same type of transition regions for yz plane **122**. Thus, for a transition region around yz plane **122** and above xz plane **120**, the processor constructs two planes making angles $\pm\theta$ with the yz plane, generally similar to planes **402** and **404**, so as to form two wedge-shaped regions terminating at the z axis and above the xz plane.

Similarly, for a transition region around yz plane **122** and below the xz plane, the processor constructs two planes making angles $\pm\theta$ with the yz plane, generally similar to planes **402** and **404**, so as to form two wedge-shaped regions terminating at the z axis and below the xz plane.

There are thus a total of four transition regions distributed symmetrically about the z-axis, each transition region comprising two wedge-shaped regions.

As for the movement for the illustrated transition region, if movement across either of planes **120** or **122** includes both wedge-shaped regions being crossed, by the HMD and the attached camera of the operator, or begins from within one of the wedge-shaped regions and crosses the other one, then the mirroring is implemented.

However, if the movement across either of the planes does not comply with the movements above, then no mirroring is implemented, i.e., mirroring is precluded.

Another disclosed embodiment of the present invention places another limitation on the mirroring described above. In this embodiment, when the operator moves to look over patient **22**, mirroring is also precluded. To preclude mirroring for this embodiment, the processor checks if the camera height, measured in step **220** of the flowchart of FIG. **7** has changed, as is the case if operator **26** moves her/his head to look over patient **22**. I.e., if the camera height changes, no mirroring is implemented regardless of whether the xz plane or the yz plane have been crossed.

FIG. **10** is a schematic illustration of the subsequent stage of the procedure, when two operators use an imaging system **320**, according to an embodiment of the present invention. Apart from the differences described below, the operation of system **320** is generally similar to that of system **20** (FIGS. **1-9**), and elements indicated by the same reference numerals in both systems **20** and **320** are generally similar in construction and in function.

In contrast to system **20**, system **320** is used by operator **26** and a second operator **326**. Second operator **326** wears an HMD **364**, and a camera **342** is fixedly attached to the HMD. HMD **364** and camera **342** are respectively substantially similar in construction and function to HMD **64** and camera **42**. However, camera **342** is typically not used to perform the registration described in the flowchart of FIG. **5**, since this is provided by camera **42**.

Images generated in HMD **364** are substantially as described in the flowchart of FIG. **7**. Thus, images presented in HMD **364** are oriented according to the point of view of camera **342**, i.e., according to whether operator **326** is to the left or right of patient **22**, and according to whether the operator is on the lower or upper side of the patient.

It will be understood that by presenting images in a head-mounted display according to the point of view of the camera attached to the display, embodiments of the present invention present correctly oriented images to a wearer of the head-mounted display. It will also be understood that the correct orientation is determined according to the position of

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the wearer of the HMD with respect to the patient, i.e., whether the wearer is to the left or right of the patient, and whether the wearer is on a lower or upper side of the patient.

It will be further understood that for cases where there is more than one HMD, each being worn by a respective wearer, embodiments of the present invention operate simultaneously and independently to present correctly oriented images to each wearer, according to the position of the respective wearer with respect to the patient. A wearer on the right side of the patient and a wearer on the left side of the patient are presented with mirror images based on anatomy images parallel to the patient sagittal plane; similarly a wearer on the lower side of the patient and a wearer on the upper side of the patient are presented with mirror images based on anatomy images parallel to the patient axial plane.

It will thus be appreciated that the embodiments described above are cited by way of example, and that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof which would occur to persons skilled in the art upon reading the foregoing description and which are not disclosed in the prior art.

The invention claimed is:

1. An imaging system, comprising:

a first head-mounted display;

a second head-mounted display;

a patient marker configured to be attached to a human subject and defining a first plane and a second plane, the patient marker comprising optically reflective elements disposed on opposing sides of the first plane in a non-symmetrical arrangement with respect to the first plane and on opposing sides of the second plane in a non-symmetrical arrangement with respect to the second plane,

wherein the patient marker is configured such that, when attached to the human subject, the first plane will make an angle between $+20^\circ$ and -20° with a sagittal plane of the human subject, and the second plane will make an angle between $+20^\circ$ and -20° with an axial plane of the human subject;

a memory configured to store a graphical representation of a tool used in a procedure performed by a wearer of the first head-mounted display on the human subject, and an image of anatomy of the human subject;

a first camera attached to the first head-mounted display and configured to acquire a first input image of the patient marker and of the tool;

a second camera attached to the second head-mounted display and configured to acquire a second input image of the patient marker and of the tool; and

one or more processors configured to:

analyze the first input image to identify the first plane and the second plane, and to identify at least one of: whether the first camera is located on a first side or second side of the first plane, or whether the first camera is located on a first side or second side of the second plane;

render to the first head-mounted display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a point of view based on the analysis of the first input image;

analyze the second input image to identify the first plane and the second plane, and to identify at least one of: whether the second camera is located on the

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first side or second side of the first plane, or whether the second camera is located on the first side or second side of the second plane; and

render to the second head-mounted display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a point of view based on the analysis of the second input image.

2. The imaging system of claim 1, wherein the first plane is orthogonal to the second plane.

3. The imaging system of claim 1, wherein the patient marker comprises a substrate, and the optically reflective elements are disposed on an upper surface of the substrate.

4. The imaging system of claim 3, wherein the patient marker further comprises a plurality of radiopaque elements disposed in the substrate on opposing sides of the first plane in a non-symmetrical arrangement with respect to the first plane and on opposing sides of the second plane in a non-symmetrical arrangement with respect to the second plane.

5. The imaging system of claim 1, wherein the image of the anatomy of the human subject is a two-dimensional image.

6. The imaging system of claim 1, wherein:

the tool comprises a reflector attached thereto,

a position of the tool in the image rendered to the first head-mounted display is based on detection of a position of the reflector in the analysis of the first input image, and

a position of the tool in the image rendered to the second head-mounted display is based on detection of a position of the reflector in the analysis of the second input image.

7. The imaging system of claim 1, wherein the one or more processors are further configured to:

analyze additional input images acquired by the first camera to identify if the first camera has moved to a different side than identified from the analysis of the first input image; and

analyze additional input images acquired by the second camera to identify if the second camera has moved to a different side than identified from the analysis of the second input image.

8. The imaging system of claim 7, wherein the one or more processors are further configured to:

detect that the first camera has moved to a different side than identified from the analysis of the first input image; and

responsive to the detection, render to the first head-mounted display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a point of view based on the detection.

9. The imaging system of claim 7, wherein the one or more processors are further configured to:

detect that the second camera has moved to a different side than identified from the analysis of the second input image; and

responsive to the detection, render to the second head-mounted display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a point of view based on the detection.

10. An imaging system, comprising:

a first head-mounted display;

a second head-mounted display;

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a first camera attached to the first head-mounted display and configured to acquire a first input image of a patient marker attached to a human subject and of a tool used in a procedure performed by a wearer of the first head-mounted display on the human subject, the patient marker comprising optically reflective elements disposed on opposing sides of a first plane in a non-symmetrical arrangement with respect to the first plane, wherein the first plane is configured to make an angle between $+20^\circ$ and -20° with a sagittal plane of the human subject, and on opposing sides of a second plane in a non-symmetrical arrangement with respect to the second plane, wherein the second plane is configured to make an angle between $+20^\circ$ and -20° with an axial plane of the human subject;

a second camera attached to the second head-mounted display and configured to acquire a second input image of the patient marker and of the tool;

a memory configured to store a graphical representation of the tool, and an image of anatomy of the human subject; and

one or more processors configured to:

analyze the first input image to identify the first plane and the second plane, and to identify at least one of: whether the first camera is located on a first side or second side of the first plane, or whether the first camera is located on a first side or second side of the second plane;

render to the first head-mounted display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a point of view based on the analysis of the first input image;

analyze the second input image to identify the first plane and the second plane, and to identify at least one of: whether the second camera is located on the first side or second side of the first plane, or whether the second camera is located on the first side or second side of the second plane; and

render to the second head-mounted display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a point of view based on the analysis of the second input image.

11. The imaging system of claim 10, wherein the first plane is orthogonal to the second plane.

12. The imaging system of claim 10, wherein the patient marker comprises a substrate, and the optically reflective elements are disposed on an upper surface of the substrate.

13. The imaging system of claim 12, wherein the patient marker further comprises a plurality of radiopaque elements disposed in the substrate on opposing sides of the first plane in a non-symmetrical arrangement with respect to the first plane and on opposing sides of the second plane in a non-symmetrical arrangement with respect to the second plane.

14. The imaging system of claim 10, wherein the image of the anatomy of the human subject is a two-dimensional image.

15. The imaging system of claim 10, wherein:

the tool comprises a reflector attached thereto,

a position of the tool in the image rendered to the first head-mounted display is based on detection of a position of the reflector in the analysis of the first input image, and

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a position of the tool in the image rendered to the second head-mounted display is based on detection of a position of the reflector in the analysis of the second input image.

16. The imaging system of claim **10**, wherein the one or more processors are further configured to:

analyze additional input images acquired by the first camera to identity if the first camera has moved to a different side than identified from the analysis of the first input image; and

analyze additional input images acquired by the second camera to identity if the second camera has moved to a different side than identified from the analysis of the second input image.

17. The imaging system of claim **16**, wherein the one or more processors are further configured to:

detect that the first camera has moved to a different side than identified from the analysis of the first input image; and

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responsive to the detection, render to the first head-mounted display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a point of view based on the detection.

18. The imaging system of claim **16**, wherein the one or more processors are further configured to:

detect that the second camera has moved to a different side than identified from the analysis of the second input image; and

responsive to the detection, render to the second head-mounted display the image of the anatomy of the human subject with the graphical representation of the tool superimposed thereon from a point of view based on the detection.

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